

right ideal (resp., finitely generated right ideal) of R is projective. Similarly, *left hereditary* and *left semihereditary* rings are defined.

2. DECOMPOSITIONS INTO PROJECTIVE AND STABLE SUBMODULES

In this section, we mention some rings over which the decomposition of some modules into projective and stable submodules occurs.

A submodule K of M is called *essential* in M , if for every submodule L of M , $L \cap K = 0$ implies $L = 0$, and it is denoted by $K \leq_e M$. A nonzero module M is said to be a *uniform* module if every nonzero submodule of M is an essential submodule. An R -module M_R is said to have *uniform dimension* (=Goldie dimension) n , denoted by $\text{u.dim}(M) = n$, where n is a positive integer, if there is an essential submodule N of M such that N is the direct sum of n (nonzero) uniform submodules. The zero module is defined to have a uniform dimension 0. If for a nonzero module M , there exists no positive integer n such that $\text{u.dim}(M) = n$, then we write $\text{u.dim}(M) = \infty$ (this will hold if and only if M contains an infinite direct sum of nonzero submodules); otherwise, we write $\text{u.dim}(M) < \infty$. A submodule K of M is said to be *small* in M if for every submodule L of M , $K + L = M$ implies $L = M$. An R -module M is said to be *hollow* (=couniform) if $M \neq 0$ and every proper submodule N of M is small in M . A finite set $\{N_i \mid i \in I\}$ of proper submodules of M is said to be *coincident* if $N_i + \left(\bigcap_{j \neq i} N_j\right) = M$ for every $i \in I$, or, equivalently, if the canonical injective mapping $M/\bigcap_{i \in I} N_i \rightarrow \bigoplus_{i \in I} M/N_i$ is bijective. An arbitrary set A of proper submodules of M is said to be *coincident* if its finite subsets are coincident. A module M is said to have finite *hollow dimension* (=couniform dimension = dual Goldie dimension) n , denoted by $\text{h.dim}(M) = n$, where n is a positive integer, if there exists a coincident set $\{N_1, N_2, \dots, N_n\}$ of proper submodules of M with M/N_i hollow for all i and $N_1 \cap N_2 \cap \dots \cap N_n$ is small in M . The zero module is defined to have a hollow dimension 0. If for a nonzero module M , there exists no positive integer n such that $\text{h.dim}(M) = n$, then we write $\text{h.dim}(M) = \infty$ (this holds if and only if there exist an infinite coincident set of proper submodules of M); otherwise, we write $\text{h.dim}(M) < \infty$. See [11, Sections 2.6, 2.7, 2.8] and [19, Section 6A].

Theorem 2.1. *If a module M cannot be decomposed as $M = P \oplus N$ where P is a projective submodule and N is a stable submodule, then there exists a sequence $(P_k)_{k=1}^\infty$ of nonzero proper projective submodules of M and a sequence $(N_k)_{k=1}^\infty$ of nonzero proper submodules of M such that for every $k \in \mathbb{Z}^+$,*

$$M = N_k \oplus P_k \oplus P_{k-1} \oplus \dots \oplus P_1 \quad \text{with} \quad N_k = N_{k+1} \oplus P_{k+1},$$

and so $\text{u.dim}(M) = \infty = \text{h.dim}(M)$.

Proof. If M were a projective module or a stable module, then it would have a decomposition of the required form trivially. So M must be a module which is not projective, and since it is not stable, it can be decomposed as $M = P \oplus N$ for some submodules P and N where P is a nonzero projective module. Then N is not projective since otherwise, $M = P \oplus N$ would be a projective module. In particular, $N \neq 0$. If N were stable, then $M = P \oplus N$ would be a decomposition into a direct sum of a projective submodule and a stable submodule. So N is not a stable module. Thus, N is neither projective nor stable. Now argue as for M . The module N should have a nonzero projective direct summand, that is, $N = P_1 \oplus N_1$ for some nonzero projective submodule P_1 and a submodule N_1 which is not projective and not stable. Continuing in this way by induction, we obtain a sequence $(P_k)_{k=1}^\infty$ of nonzero proper projective submodules of M and a sequence $(N_k)_{k=1}^\infty$ of nonzero proper submodules of M such that

$$M = N_k \oplus P_k \oplus P_{k-1} \oplus \dots \oplus P_1 \quad \text{with} \quad N_k = N_{k+1} \oplus P_{k+1} \quad \text{for all} \quad k \in \mathbb{Z}^+.$$

Since $P_i \neq 0$ for all $i \in \mathbb{Z}^+$, $\text{u.dim}(P_i) \geq 1$ and so for every $n \in \mathbb{Z}^+$, $\text{u.dim}(M) = \text{u.dim}(N_n \oplus P_n \oplus P_{n-1} \oplus \dots \oplus P_1) \geq n$ by [19, 6.6]. Therefore $\text{u.dim}(M) = \infty$. Similarly $\text{h.dim}(M) = \infty$ should hold using the properties of hollow dimension, see [11, Section 2.8] $\square$